

A JEPA-LLM Tuning Framework for Self-Supervised Medical Image Segmentation and DNA Fingerprinting Analysis

TECHNICAL SPECIFICATION & ABSTRACT

1. Abstract

This document details a unified architectural framework combining Joint Embedding Predictive Architecture (I-JEPA/V-JEPA) concepts with Large Language Model (LLM) fine-tuning strategies to solve two critical computational biomedicine challenges: high-fidelity Self-Supervised Medical Image Segmentation and high-throughput DNA Fingerprinting Analysis. By shifting away from pixel-level/token-level generative reconstruction toward abstraction-level semantic prediction in a joint embedding space, the framework optimizes representation learning without heavy pixel-level decoders. The framework incorporates a multi-modal LLM alignment bridge that converts abstract vision embeddings and sequence-based DNA representations into structured instructions. This allows the model to map diverse cellular structures and non-coding genomic repeat sequences onto precise phenotypic and structural boundaries. The resulting architecture drastically reduces annotated sample complexity, preserves geometric topological structures across radiological/pathological imaging modalities, and enhances genomic variant verification.

2. Key Architectural Features

2.1 Non-Generative Self-Supervised Joint Embedding (JEPA-Core)

Unlike traditional Masked Autoencoders (MAE) or contrastive frameworks, the JEPA core eliminates pixel-level and nucleotide-level generation. By projecting target images and sequence windows into an abstracted semantic space, the network uses a context encoder to predict the representation of a target block from a context block. This completely circumvents computational overhead on pixel noise, focusing entirely on high-level topological continuity in medical images and structural repeat regularities in genomic sequences.

2.2 Multi-Scale Cross-Attention Contextual Predictor

The framework utilizes a specialized predictor network utilizing multi-scale cross-attention mechanisms. For medical image segmentation, this enables the tracking of macro-anatomical variations alongside micro-cellular margins. For DNA fingerprinting, it tracks Short Tandem Repeats (STRs) and Variable Number Tandem Repeats (VNTRs) by treating multi-scale sequence lengths as dynamic prediction windows in the latent domain.

2.3 LLM Tuning & Modality Alignment Bridge

To leverage the zero-shot reasoning capabilities of LLMs, the framework binds the learned JEPA representations via a lightweight alignment block (Perceiver Resampler or Q-Former). The LLM is fine-tuned using Parameter-Efficient Fine-Tuning (PEFT) methods like Low-Rank Adaptation (LoRA). This alignment allows the model to process specialized visual/genomic tokens and generate cross-modal reasoning paths—translating a segmented tumor boundary or an STR allele match into cross-referenced, clinically structured text and coordinates.

2.4 Dual-Domain Spatial & Sequence Encoding

The framework maps heterogeneous inputs into a unified token stream using a dual-domain tokenization layer:

- **Spatial Domain:** ViT-based patch tokenization with 2D/3D sinusoidal position embeddings to preserve anatomical coordinates in MRIs, CT scans, and histopathology.
- **Sequence Domain:** K-mer tokenization paired with relative position encodings to maintain the exact spatial proximity of hypervariable genomic loci.

2.5 Low-Resource Sample Efficiency & Label Minimalist Adaptation

By leveraging self-supervised pre-training via the predictive embedding space, the model overcomes the data scarcity bottleneck inherent to specialized medicine. The learned representations can successfully fine-tune for complex multi-organ segmentation or forensic DNA match profiling with fewer than 5% of traditional dense pixel-level or token-level labels.

3. Mathematical Formulation Overview

The training objective eliminates direct pixel or token reconstruction loss. Let $\mathbf{x}_c$ be the context patch and $\mathbf{x}_t$ be the target patch. The context encoder E_c and target encoder E_t map inputs to representations. The predictor P forecasts the target representation based on target mask coordinates Δ_t :

$$L_{\{JEPA\}} = \| P(E_c(\mathbf{x}_c), \Delta_t) - E_t(\mathbf{x}_t) \|^2$$

This is combined with the autoregressive LLM alignment loss during the multi-modal tuning phase to refine boundary coordinates and sequence classification tags simultaneously.